

Brain and Behavioral Imaging Analysis Workflow

Brain Recording Workflow

1. Convert .nd2 Files to .h5

- Use scripts in Cf0-TreatND2 to convert raw .nd2 files to .h5 format.

2. Coarse Segmentation and Alignment

- Open the .h5 file in the GUI.
- Perform coarse segmentation on all frames.
- Choose one reference frame for alignment (see *Targettrack* paper).
- Align all frames to the reference frame.
- Crop the reference frame.
- Export:
 - Go to Import/Export tab in GUI.
 - Click “Export Cropped and Rotated”.
 - This saves both channels in a smaller, cropped and rotated format.

3. Export Red Channel

- Open the cropped and rotated file.
- Export the red channel only for an even smaller, easier-to-use file (*).

4. Annotation and Neural Network Training

- Annotate a few frames from the (*) red channel file.
- Train the neural network on these frames to get predictions.
- Approve prediction masks.
- Manually correct masks in the Annotation tab.

- Optional:
 - Use point tracking for neurons with nucleic expressions.
 - Use scripts in `Cf1-PointData` to convert masks to points and vice versa.

5. Neural Network Training on HPC Cluster

- Use `Cf2-TransferToCluster_to_train_h5_remotely` scripts.
- Note: Targettrack's targeted augmentation doesn't currently work on Kuma cluster because Jian-Vermuri point registration package is not updated for versions of python and numpy used on cluster. Consider using normal rotations.
- Commands:

- connect to cluster

```
ssh barzegar@kuma.hpc.epfl.ch
```

- Submit a job to for training point tracking

```
sbatch runArrayPts.sbatch Movie.h5
```

- Submit a job to for training mask tracking

```
sbatch runArraySingle.sbatch 53 1 SJR16Sep20
```

- Check the status of the job

```
squeue -u barzegar
```

- Transfer the results from cluster into local folders

```
scp -r barzegar@kuma.hpc.epfl.ch:/scratch/barzegar/  
neural_network_scripts_2024/movie.h5 / Your/Local/Directory/
```

6. Rotation Corrections (Post-training)

- If initial rotations are poor, use `Cf1-roatationScripts/`.
- Rotate frames based on 2 well-tracked neurons.

7. Activity Extraction

- Map masks from red channel (*) back to the cropped or original .h5 file.
- Use `Cf3-ExtractActivity/` scripts to extract calcium activity.

Behavioral Recording Workflow

1. Initial Coarse Annotation

- Previously: Used DeepLabCut + manual correction.
- Now: Use SAM or Mediar.

2. Cropping

- Use `B0_PreProcess_forSAM/0-Cropavi.ipynb` to crop the field-of-view to the microfluidic chip.

3. Partitioning Large .avi Files

- Use `B0_PreProcess_forSAM/1-PreProcess_files.ipynb`.
- Split files to 30–50MB before using SAM.

4. Segmentation with SAM

- Use modified SAM in `LPBS2/Misc/Mahsa-Barzegar/Revisions/ExtraData/SAMPT`.
- Optional: Replace bad first frames using `AddFr_toVideo.ipynb`.

5. Mask Collection and Skeletonization

- Collect masks using `0-CollectMasks.ipynb`.
- Skeletonize masks using `skletonize.ipynb` to get 200-point coordinates.

6. Convert Skeletons to DLC Format

- Use `2-ImportMasksToDLC_0319.ipynb`.
- Manually correct masks in DLC.

Joystick and Camera Alignments

Joystick Correction

- Run `0-AlignUppeCamera_Joystick_diffDelays.ipynb`.
- Use the output in `1-AlignUppeCamera_Joystick.ipynb`.

Upper and Lower Camera Alignment

- Use `AlignUpperandLowerCamera.ipynb`.